# Is merging worth it? Securely evaluating the information gain for causal dataset acquisition

The repository of the source code mentions the experiment folders and the main logic-based Python files (*.py*) or Jupyter Notebooks (*.ipynb*) separately. It also contains spreadsheets (*.csv*) for the datasets and other YAML files. Ease in reproducibility has been enhanced by organizing the repo as per the sections in the paper. (see README file). The Jupyter Notebooks are helpful very interactive as they consist of the input, the output and the plots. Due to the absence of certain build tools, running *pip* command for certain modules on Windows is a headache. Qualitative trends can be reproduced reliably; exact numeric values may differ unless seeds and library versions are controlled. Sadly, the README file does not mention the runtime estimates and hardware notes. Also, the file does not mention the suitable Python versions for the libraries used. For more details, see the following list.

1) **Repository**
    a) Repository link
    b) README file
    c) requirements.txt

2) **Dependencies**
    a) Python 3.10 or 3.11
    b) OS:
        i) Recommended: Linux (WSL2) with conda-forge
        ii) Available: Windows 11 (no build isolation) with Visual Studio Build Tools (MSVC)
    c) Libraries (see the requirements.txt file in the repository for full environment details)
        i) NumPy
        For most of the modules (e.g., xbcausalforest) having NumPy, *pip* creates an isolated build environment during the build. Most often, that environment does not include NumPy, even though NumPy is installed globally; thence showing:

        *ModuleNotFoundError: No module named 'numpy'*

ii) pandas
iii) sys
iv) matplotlib
v) tqdm
vi) torch

***UserWarning**: torch.set_default_tensor_type() is deprecated as of PyTorch 2.1, please use **torch.set_default_dtype()** and **torch.set_default_device()** as alternatives.*

vii) os
viii) random
ix) datetime
x) scipy
xi) Time

| **Deprecated Packages** | **Packages for pip commands** |
|---|---|
| xii) sklearn | scikit-learn |
| xiii) yaml | PyYAML |
| xiv) pyro | pyro-ppl |

| **Packages** | **Isolated Build Environment** | **Local C++ Compilation (on Windows)** |
|---|---|---|
| xv) GPy (only supported on Python 3.9 - 3.12) | Required | Not Required |
| xvi) xbcausalforest | Required | Required |
| xvii) xbart | Required | Required |
| xviii) cmgp | Required | Required |

3) **Hardware** (used in reproduction of src code)
   *a)* CPU: AMD Ryzen 5 7520U with Radeon Graphics (2.80 GHz) *(sufficient for moderate workloads)*
   b) Installed RAM: 16.0 GB
   c) System Type: 64-bit operating system, x64-based processor
   d) GPU: AMD Radeon(TM) 610(M) *(no official ROCm support)*
   e) TPU: Not Available
   f) Parallel Computing Capability: 4 Cores *(optimal performance)*